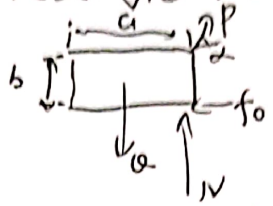
CODE OF STUDENT 2357

Number of Problem: 5-1

Solutions: For the rectangle we can draw the force analysis



Case 1: we can write the force balance equations:

in x direction: $P\cos\alpha = f_0$. ... ①

in y direction: $P\sin\alpha + N = Q$. ... ②

without movement: $\frac{f_0}{N} \le f$.

we can get that

$$P \le \frac{fQ}{\cos\alpha + f\sin\alpha}.$$

To ensure the equilibrium. we solve the minimum of the function:

$$f(\alpha) = \frac{fQ}{\cos\alpha + f\sin\alpha}$$

And get

$$f(\alpha)_{min} = \frac{fQ}{\sqrt{1+f^2}}.$$

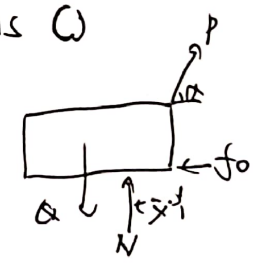
so the force P must meet

$$P \le \frac{fQ}{\sqrt{1+f^2}}$$

Case 2: The rectangle without rotation:

we can also write the equation: (the center is C)

[The distance x is shown in the figure]



$$Q\cdot\frac{a}{2} = P\cdot\cos\alpha\cdot b + N\cdot x \quad \text{--- ③}$$

For the equation ② we can get

$$N = Q - P\sin\alpha$$

In equation ③ we can get

$$x = \frac{Q\cdot\frac{a}{2} - Pb\cos\alpha}{Q - P\sin\alpha} \ge 0.$$

And

$$P \le \frac{Q\cdot\frac{a}{2}}{b\cos\alpha}$$

Similar to case 1. we can get the minimum of the function:

$$P \le \frac{Qa}{2b}.$$

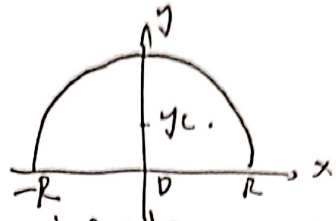
All in all. the force P should satisfy:

$$P \le \min\left\{\frac{fQ}{\sqrt{1+f^2}}, \frac{Qa}{2b}\right\}.$$

CODE OF STUDENT 2 3 7

Number of Problem: 5-2

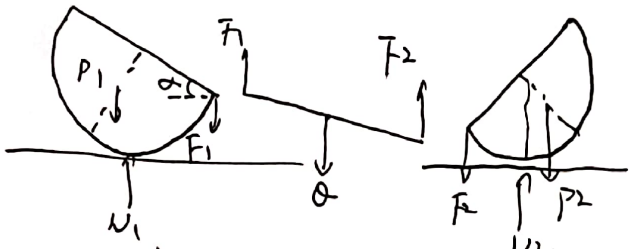
Solutions: First we must solve the center of mass of the half-disc



$$y_c = \frac{\int_0^{\pi} d\theta \int_0^R \sigma \cdot r\sin\theta \cdot r\,dr}{\int_0^{\pi} d\theta \int_0^R \sigma \cdot r\,dr\,d\theta} = \frac{\sigma \cdot \frac{2}{3}R^3}{\pi\sigma\frac{R^2}{2}} = \frac{4R}{3\pi}.$$

And since the floor is smooth. The force in the hinges are just in the vertical direction:

So. we can draw the force analysis:



For the rod, we can solve that

$$F_1 = F_2 = \frac{Q}{2}.$$

~~$P_1 = P_2 = \frac{Q}{2}$~~

For the half-~~disc~~ discs, in order to hold the balance:

$$P_1 \cdot \frac{4R_1}{3\pi}\sin\alpha_1 = F_1 \cdot R_1 \cdot \cos\alpha_1$$

$$P_2 \cdot \frac{4R_2}{3\pi} \cdot \sin\beta = F_2 \cdot R_2 \cdot \sin\beta.$$

We can solve that

$$\tan\alpha = \frac{3\pi Q}{2P_1}$$

$$\tan\beta = \frac{3\pi Q}{2P_2}.$$

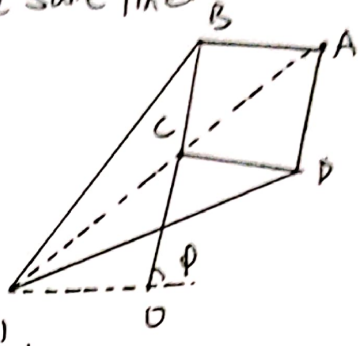
So the angles are

$$\alpha = \tan^{-1}\left(\frac{3\pi Q}{2P_1}\right). \quad \beta = \tan^{-1}\left(\frac{3\pi Q}{2P_2}\right).$$

CODE OF STUDENT 2 3 7

Number of Problem: k-1

Solutions: According to the relations in the figure. we can know that point O_1. C. A are on the same line
And $\angle BO_1C = \angle DO_1C$.



For $O_1O = OC = r$. then

$\angle CO_1O = \frac{\varphi}{2}$.

Then $OK = 2r\cos\frac{\varphi}{2}$.

in $\triangle BCO_1$. we can get

$$BC^2 + CO_1^2 - 2BC \cdot CO_1 \cdot \cos\angle BCO_1 = BO_1^2$$

$$\cos\angle BCO_1 = \frac{BC^2 + CO_1^2 - BO_1^2}{2BC \cdot CO_1}$$

$$= \frac{a^2 + (2r\cos\frac{\varphi}{2})^2 - l^2}{2a \cdot 2r\cos\frac{\varphi}{2}}$$

For $\angle BCO_1 + \angle BCA = \pi$.

$\cos\angle BCO_1 + \cos\angle BCA = 0$.

And $AC = 2a\cos\angle BCA$

we can get the length of O_1A:

$$O_1A = 2r\cos\frac{\varphi}{2} + \frac{l^2 - a^2 - (2r\cos\frac{\varphi}{2})^2}{2r\cos\frac{\varphi}{2}}$$

So. the coordination of A can described as (r, θ).

$$\begin{cases} r = 2r\cos\frac{\varphi}{2} + \frac{l^2 - a^2 - (2r\cos\frac{\varphi}{2})^2}{2r\cos\frac{\varphi}{2}} \\ \theta = \frac{\varphi}{2}. \end{cases}$$

we can simplify the formula of r:

$$r = \frac{l^2 - a^2}{2r\cos\frac{\varphi}{2}}. \qquad \theta = \frac{\varphi}{2}.$$

For $\begin{cases} r = r(t) \\ \theta = \theta(t) \end{cases}$, we can get the velocity formula:

$$\vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta}$$

And the acceleration formula:

$$\vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta}$$

For $\begin{cases} r = \frac{l^2 - a^2}{2r\cos\frac{\varphi}{2}} \\ \theta = \frac{\varphi}{2} \end{cases}$. we can calculate that $[\dot{\varphi} = \omega.\ \ddot{\varphi} = 0]$

CODE OF STUDENT 237

Number of Problem: 4-1

$$\begin{cases} \dot{r} = \dfrac{(l^2-a^2)2r}{(2r\cos\frac{\varphi}{2})^2}\sin\dfrac{\varphi}{2}\cdot\dfrac{\omega}{2} \\ \dot{\theta} = \dfrac{\omega}{2}. \end{cases}$$

So: $\vec{V} = \left[\dfrac{(l^2-a^2)2r}{(2r\cos\frac{\varphi}{2})^2}\cdot\sin\dfrac{\varphi}{2}\cdot\dfrac{\omega}{2}\right]\hat{r} + \left[r\cdot\dfrac{\omega}{2}\right]\hat{\theta}$

And $\begin{cases} \ddot{r} = \dfrac{(l^2-a^2)\omega^2}{4r}\cdot\dfrac{1-\frac{1}{2}\cos^2\frac{\varphi}{2}}{\cos^3\frac{\varphi}{2}} \\ \ddot{\theta} = 0. \end{cases}$

So $\vec{a} = \left[\dfrac{(l^2-a^2)\omega^2}{4r}\cdot\dfrac{1-\frac{1}{2}\cos^2\frac{\varphi}{2}}{\cos^3\frac{\varphi}{2}} - r\cdot\dfrac{\omega^2}{4}\right]\hat{r}$

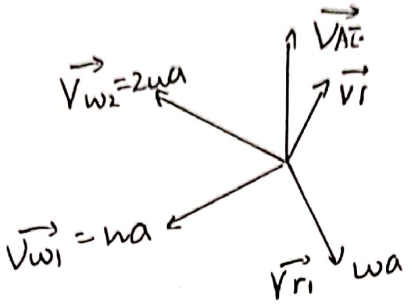
$+ \left[2\cdot\dfrac{\omega}{2}\cdot\dfrac{(l^2-a^2)2r}{(2r\cos\frac{\varphi}{2})^2}\sin\dfrac{\varphi}{2}\cdot\dfrac{\omega}{2}\right]\hat{\theta}.$

For $\hat{r}$ is in $\vec{OA}$ direction. $\hat{\theta}\perp\hat{R}$.

CODE OF STUDENT 237

Number of Problem: k-2

Solution: when the triangle EDE is equilateral. we can draw the velocity analysis:



The velocity of the slider E. relatively to the rod BC is $\vec{V}_r$.

$$\vec{V}_{w2}+\vec{V}_r=\vec{V}_{w1}+\vec{V}_{r1}+\vec{V}_{AE}.$$

In $\vec{V}_{w2}$ direction:

$$\frac{1}{2}wa+\frac{1}{2}V_{AE}-\frac{\sqrt{3}}{1}wa=2wa$$

$$V_{AE}=(3+\sqrt{3})wa.$$

In $\vec{V}_r$ direction:

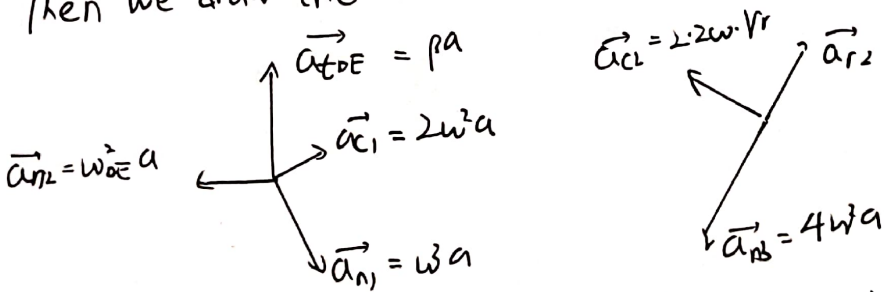
$$V_r=\frac{-\sqrt{3}}{2}wa-\frac{1}{2}wa+\frac{\sqrt{3}}{2}V_{AE}.$$

$$V_r=(\sqrt{3}+1)wa.$$

the angular velocity of the rod DE is:

$$w_{DE}=\frac{V_{DE}}{a}=(3+\sqrt{3})w$$

Then we draw the acceleration analysis:



$$\vec{a}_{n2}+\vec{a}_{n1}+\vec{a}_{tDE}+\vec{a}_{c1}=\vec{a}_{r0}+\vec{a}_0+\vec{a}_{r2}.$$

in $\vec{a}_0$ direction:

$$a_{c2}=\frac{\sqrt{3}}{2}a_{n2}+\frac{1}{2}a_{tDE}-\frac{1}{2}a_{c1}-\frac{\sqrt{3}}{2}a_{n1}$$

$$4(\sqrt{3}+1)w^2a=\frac{\sqrt{3}}{2}(3+\sqrt{3})^2w^2a+\frac{1}{2}a_{tDE}-\frac{1}{2}\cdot 2w^2a-\frac{\sqrt{3}}{2}w^2a$$

$$a_{tDE}=-(8+3\sqrt{3})w^2a.$$

Number of Problem: k-2

CODE OF STUDENT 237

And in $\vec{a_{r_2}}$ direction:

$$a_{r_2} - 4w^2a = \frac{\sqrt{3}}{2} a_{DE} + \frac{\sqrt{3}}{2} a_{C1} - \frac{1}{2} a_{n_2} - \frac{1}{2} a_{n_1}$$

$$= \frac{\sqrt{3}}{2}(6+3\sqrt{3})w^2a + \frac{\sqrt{3}}{2} \cdot 2w^2a - \frac{1}{2} \cdot (3+\sqrt{3})^2 w^2 a - \frac{1}{2} \cdot w^2 a$$

we can get $a_{r_2} = -(7+6\sqrt{3})w^2a$

So the velocity, acceleration E relatively to the rod BC is:

$$v_r = (\sqrt{3}+1)wa \qquad \vec{a_{r_2}} = -(7+6\sqrt{3})w^2a.$$

The angular velocity of the rod DE is

$$w_{DE} = (3+\sqrt{3})w.$$

CODE OF STUDENT 237

Number of Problem: P-1

Solution: we can draw the force analysis:



The functions:

$$\begin{cases} T\cos\alpha + N = mg \\ T\sin\alpha = m\dfrac{v^2}{l\sin\alpha} \\ d_0 = fN = -m\dot{v} \end{cases}$$

And we can get:

$$f\left(mg - \frac{mv^2\cos\alpha}{l\sin\alpha}\right) = -m\frac{dv}{dt}.$$

For the initial velocity $V|_{t=0} = v_0$. $\quad V|_{x=\pi l\sin\alpha} = 0.$

To ensure no detach of the point from the plane.

$$v = 0. \quad t = 0.$$

so we can get

$$T \geq \frac{mg}{\cos\alpha}$$

The minimum is $T = \frac{mg}{\cos\alpha}$

And $\quad T\sin\alpha = m\frac{v^2}{l\sin\alpha}$

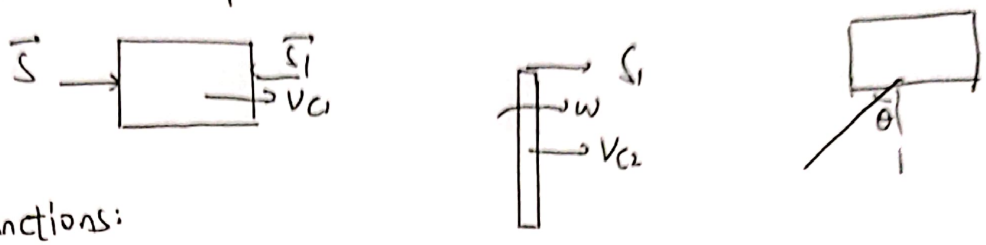
so $\quad v = \sqrt{gl\frac{\sin^2\alpha}{\cos\alpha}}$

The minimum initial velocity is $v_{min} = \sqrt{gl\frac{\sin^2\alpha}{\cos\alpha}}$

CODE OF STUDENT 2377

Number of Problem: D-2

Solutions: we can describe the impulse between the slide and the rod is S_1. Then:



The functions:

$$\begin{cases} S - S_1 = m_1 v_1 \\ S_1 = m_2 v_{C2} \\ S_1 \cdot \frac{l}{2} = \frac{1}{12} m_2 l^2 \cdot w. \end{cases}$$

And the relations in velocity:

$$v_1 = v_{C2} + \frac{l}{2} w.$$

we can solve that:

$$S_1 = \frac{m_2}{4m_1 + m_2} S \qquad v_{C2} = \frac{S}{4m_1 + m_2}$$

$$v_1 = \frac{4S}{4m_1 + m_2} \qquad w = \frac{6S}{(4m_1 + m_2) l}$$

And the velocity of the centre of the mass m_1 and m_2 is:

$$v_C = \frac{S}{m_1 + m_2}$$

The dynamics energy of m_1 and m_2 is:

$$\bar{E}_{K1} = \frac{1}{2} m_1 v_1^2$$

$$E_{kC2} = \frac{1}{2} m_2 v_{C2}^2 + \frac{1}{2} \left(\frac{1}{12} m_2 l^2\right) \cdot w^2$$

So. we can get

$$\bar{E}_K = \bar{E}_{K1} + \bar{E}_{K2} = \frac{(8m_1 + 2m_2) S^2}{(4m_1 + m_2)^2}$$

The energy is consist of the dynamics energy of the centre of mass, and the $\bar{E}_P$ of m_2. (when arrive at the maxinal of the angle of the rod).

$$E_K = E_{KC} + \bar{E}_P. \quad \text{(use concenation of energy).}$$

$$\bar{E}_{KC} = \frac{1}{2} (m_1 + m_2) v_C^2 = \frac{1}{2} \frac{S^2}{(m_1 + m_2)}$$

$$\bar{E}_P = m_2 g \frac{l}{2} (1 - \cos\theta).$$

So. we can get.

Number of Problem: D-2

CODE OF STUDENT 2307

$$\bar{E}_p = \bar{E}_{1c} - \bar{E}_{fc} = \frac{(8m_1+2m_2)s^2}{(4m_1+m_2)^2} - \frac{s^2}{2(m_1+m_2)} = \frac{3m_2 s^2}{2(m_1+m_2)(4m_1+m_2)}$$

And finally we can solve that,

$$\cos\theta = 1 - \frac{3s^2}{(m_1+m_2)(4m_1+m_2)gl}$$

if $1 - \frac{3s^2}{(m_1+m_2)(4m_1+m_2)gl} < -1$. then

the maximal deflection angle is π.